QuickPhone — Physical ERD (complete): 16 tables with all keys and attributes

One diagram for the whole data model. Each line is named both ways: top or left from the "one" side, bottom or right back from the "many" side. PK primary key · FK foreign key (→ table) · UK unique · CK checked value list · NN not null · derived = denormalized

store — D7 Staff & Stores

PK	**store_id**	INTEGER	NN
UK	store_name	VARCHAR(60)	NN
	address	VARCHAR(200)	NN
	phone	VARCHAR(20)	NN
	email_address	VARCHAR(120)	NN
	business_hours	VARCHAR(120)	

appointment — D2 Appointments

PK	**appointment_id**	INTEGER	NN
FK	customer_id	INTEGER	→ *customer*
FK	store_id	INTEGER	→ *store*
FK	technician_id	INTEGER	→ *employee*
	scheduled_at	TIMESTAMP	NN
CK	appt_status	VARCHAR(12)	NN

repair_job — D3 Repair Jobs

PK	**repair_job_id**	INTEGER	NN
UK	ticket_number	VARCHAR(12)	NN
FK	device_id	INTEGER	→ *device*
FK	store_id	INTEGER	→ *store*
FK,UK	appointment_id	INTEGER	→ *appointment*
FK	assigned_tech_id	INTEGER	→ *employee*
CK	status	VARCHAR(20)	derived · NN
	reported_issue	VARCHAR(500)	NN
	intake_condition	VARCHAR(500)	
	diagnosis_notes	VARCHAR(2000)	
	repair_notes	VARCHAR(2000)	
	check_in_at	TIMESTAMP	NN
	completed_at	TIMESTAMP	
	final_cost	DECIMAL(10,2)	
	email_sent_at	TIMESTAMP	
FK	email_sent_by_id	INTEGER	→ *employee*
CK	email_status	VARCHAR(10)	
	picked_up_at	TIMESTAMP	
	id_verified_flag	BOOLEAN	NN
	signature_file	VARCHAR(255)	

payment — D3 Repair Jobs

PK	**payment_id**	INTEGER	NN
FK	repair_job_id	INTEGER	→ *repair_job*
	amount	DECIMAL(10,2)	NN
CK	method	VARCHAR(12)	NN
CK	payment_status	VARCHAR(10)	NN
	paid_at	TIMESTAMP	NN

employee — D7 Staff & Stores

PK	**employee_id**	INTEGER	NN
FK	store_id	INTEGER	→ *store*
	first_name	VARCHAR(40)	NN
	last_name	VARCHAR(40)	NN
UK	email	VARCHAR(120)	NN
CK	role	VARCHAR(12)	NN
	password_hash	VARCHAR(255)	NN
	active_flag	BOOLEAN	NN

customer — D1 Customers & Devices

PK	**customer_id**	INTEGER	NN
	first_name	VARCHAR(40)	NN
	last_name	VARCHAR(40)	NN
IX	phone	VARCHAR(20)	NN
	email	VARCHAR(120)	
	created_at	TIMESTAMP	NN

job_part — D3 Repair Jobs

PK	**job_part_id**	INTEGER	NN
FK	repair_job_id	INTEGER	→ *repair_job*
FK	part_id	INTEGER	→ *part*
FK	purchase_order_id	INTEGER	→ *purchase_order*
	quantity	INTEGER	NN
CK	part_status	VARCHAR(8)	NN
	recorded_at	TIMESTAMP	NN

device — D1 Customers & Devices

PK	**device_id**	INTEGER	NN
FK	customer_id	INTEGER	→ *customer*
	device_type	VARCHAR(20)	NN
	brand	VARCHAR(40)	NN
	model	VARCHAR(60)	NN
IX	serial_or_imei	VARCHAR(40)	

job_status_history — D3 Repair Jobs

PK	**history_id**	BIGINT	NN
FK	repair_job_id	INTEGER	→ *repair_job*
CK	status	VARCHAR(20)	NN
FK	changed_by_id	INTEGER	→ *employee*
	changed_at	TIMESTAMP	NN
	note	VARCHAR(300)	

part — D4 Parts Inventory

PK	**part_id**	INTEGER	NN
FK	default_supplier_id	INTEGER	→ *supplier*
	part_name	VARCHAR(100)	NN
	device_model	VARCHAR(60)	
	grade	VARCHAR(20)	
	standard_cost	DECIMAL(10,2)	NN
	active_flag	BOOLEAN	NN

supplier — D6 Suppliers

PK	**supplier_id**	INTEGER	NN
UK	supplier_name	VARCHAR(100)	NN
	contact_name	VARCHAR(80)	
	phone	VARCHAR(20)	
	email	VARCHAR(120)	
	active_flag	BOOLEAN	NN

store_part_stock — D4 Parts Inventory

PK,FK	**store_id**	INTEGER	→ *store*
PK,FK	**part_id**	INTEGER	→ *part*
CK	quantity_on_hand	INTEGER	derived · NN
	reorder_point	INTEGER	NN

quote — D3 Repair Jobs

PK	**quote_id**	INTEGER	NN
FK	repair_job_id	INTEGER	→ *repair_job*
	labor_amount	DECIMAL(10,2)	NN
	parts_amount	DECIMAL(10,2)	NN
	total_amount	DECIMAL(10,2)	derived · NN
CK	decision	VARCHAR(10)	NN
	decision_at	TIMESTAMP	
FK	decision_by_id	INTEGER	→ *employee*

inventory_txn — D4 Parts Inventory

PK	**inventory_txn_id**	BIGINT	NN
FK	store_id	INTEGER	→ *store_part_stock*
FK	part_id	INTEGER	→ *store_part_stock*
FK	recorded_by_id	INTEGER	→ *employee*
CK	txn_type	VARCHAR(12)	NN
	quantity_change	INTEGER	NN
	reason	VARCHAR(200)	
	txn_at	TIMESTAMP	NN

purchase_order — D5 Purchase Orders

PK	**purchase_order_id**	INTEGER	NN
FK	store_id	INTEGER	→ *store*
FK	supplier_id	INTEGER	→ *supplier*
FK	created_by_id	INTEGER	→ *employee*
FK	approved_by_id	INTEGER	→ *employee*
CK	po_status	VARCHAR(20)	NN
	created_at	TIMESTAMP	NN
	approved_at	TIMESTAMP	
	placed_at	TIMESTAMP	
	rejection_reason	VARCHAR(300)	

po_line — D5 Purchase Orders

PK	**po_line_id**	INTEGER	NN
FK	purchase_order_id	INTEGER	→ *purchase_order*
FK	part_id	INTEGER	→ *part*
	quantity_ordered	INTEGER	NN
	quantity_received	INTEGER	NN
	unit_cost	DECIMAL(10,2)	NN

schedules / is scheduled at — store ↔ appointment

employs / is employed by — store ↔ employee

handles / is handled by — employee ↔ appointment

books / is booked by — customer ↔ appointment

leads to / results from — appointment ↔ repair_job

works on / is assigned to — employee ↔ repair_job

sends notification for / has notification sent by — employee ↔ repair_job

receives / is received for — repair_job ↔ payment

uses / is used on — repair_job ↔ job_part

is included in / includes — job_part ↔ part

owns / is owned by — customer ↔ device

undergoes / is performed on — device ↔ repair_job

tracks / belongs to — repair_job ↔ job_status_history

generates / is generated for — employee ↔ job_status_history

appears in / is stock of — part ↔ store_part_stock

logs / is logged for — store_part_stock ↔ inventory_txn

creates / is created by — employee ↔ purchase_order

fulfills / is fulfilled by — supplier ↔ purchase_order

approves / is approved by — employee ↔ purchase_order

provides / is provided by — supplier ↔ part

changes / is made by — employee ↔ job_status_history

decides / is decided by — employee ↔ quote

contains / is contained in — purchase_order ↔ po_line

records / is recorded by — employee ↔ inventory_txn

appears on / lists — part ↔ po_line

supplies / is supplied by — supplier ↔ po_line

Every foreign key is drawn as a relationship line. EMPLOYEE plays several roles: the technician assigned a job, the sender of the pickup notification, and the person who changes a status, decides a quote, records a stock change, or creates or approves a purchase order.

QuickPhone Repair Tracking System: Physical ERD notes

MIS 4173 Sprint 3 · Team 4 · Method: Dennis, Wixom, Roth & Haley, *Systems Analysis and Design*, 8th ed., Ch. 5 and Ch. 10

Logical to physical (Fig. 10-7)

1. **Entities to tables.** Sixteen tables in snake_case; each header names the DFD data store (D1–D7) that holds it, so the ERD and the DFDs stay balanced.
2. **Attributes to fields.** Each field has a data type and size (VARCHAR, INTEGER, DECIMAL, TIMESTAMP, BOOLEAN), NN for not null, and CK where only listed values are allowed (statuses, roles, payment method). Each field's default and valid values (Fig. 10-9) are in the models document, Appendix A.
3. **Primary keys.** Every table has one (PK); store_part_stock uses the pair store_id + part_id.
4. **Foreign keys.** Every relationship is a foreign key in the child table (FK, with the parent table named). Every foreign key is drawn as a line with crow's-foot cardinality, and each line is named both ways, as in Fig. 10-8 ("Places / Is Placed By"): the first name reads from the "one" side to the "many" side, the second reads back (Ch. 5).
5. **System-related components.** Each field the physical DFD adds is stored where its flow writes it: password_hash by 9.1 Maintain Staff Accounts (Staff Account Input Form); ticket_number by 2.4 Create Repair Job, printed by 2.5 on the Claim Ticket and Device Tag Printouts; email_sent_at, email_sent_by_id and email_status by 5.5 Record Notification Result (Email Send Result); id_verified_flag by 6.2 Record Identity Check (ID Check Input Form); signature_file and picked_up_at by 6.4 Close Repair Job (Signature Input Form). The audit tables take a row from every status change (job_status_history) and from 4.3, 7.5, 9.3 and the 9.5 Inventory Import File batch (inventory_txn); created_at and the other time stamps are set by the application on save. The Import Error Report goes back to the Manager and is not stored.

Documented denormalizations

The logical ERD is in third normal form. Three fields marked *derived* are stored on purpose for speed and are kept correct by the program (Ch. 10, optimizing storage):

- `repair_job.status`: the latest row of job_status_history.
- `quote.total_amount`: labor_amount + parts_amount.
- `store_part_stock.quantity_on_hand`: the sum of inventory_txn.quantity_change.

Assumptions

Types are standard SQL, all valid in PostgreSQL, the chosen DBMS. DEVICE stays its own table inside D1: storage is optimized table by table (no repeated customer data, few nulls; Ch. 10), so a separate data store would save nothing. EMPLOYEE.store_id is a home store; the Owner sees both stores. Photo ID numbers are never stored, only that the check was done.